

The solution of the recurrence relation $a_n = 2a_{n-1}$ with $a_0 = 1$ is

- a) $a_n = 2^n + 2^{n-1}$ b) $a_n = 2^{n+1}$
 c) $a_n = 2^{n-1}$ d) $a_n = 2^n$

The generating function for the sequence 2,4,1,1,1,1,... is

- a) $1 + 3x + \frac{1}{1-x}$ b) $1 + 3x + \frac{1}{1+x}$
 c) $2 + 4x + \frac{1}{1-x}$ d) $2 + 4x + \frac{1}{1+x}$

Let p be “today is Sunday” and q be “the time is 4:00 PM”, then the statement “it is false that today is Sunday or time is 4:00 PM” is logically equivalent to

- a) $\neg(p \vee q)$ b) $\neg(\neg p \vee q)$
 c) $\neg(\neg(p \vee q))$ d) $\neg(p \vee \neg q)$

Answer:a)

The degree of recurrence relation $a_{r-2} + 2a_{r-3} + 3a_{r-5} = 1$ is

- a) 2 b) 3
 c) 4 d) 1

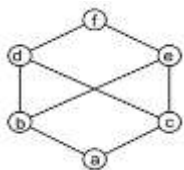
Answer:b)

Which of following represents a tautology?

- a) $p \wedge \neg p$ b) $\neg p \wedge \neg p$
 c) $p \vee \neg p$ d) None of them

Answer: c)

The sum of degree of all the vertices in below graph is



- a) 12 b) 15
 c) 16 d) 18

Answer: c)

A self-complemented distributive lattice is called

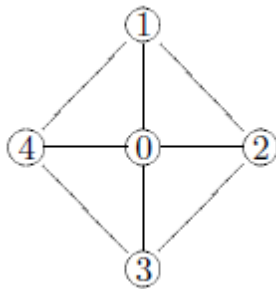
- a) Modular lattice
- b) Boolean algebra
- c) Complete lattice
- d) Self dual lattice

Answer (b)

If there is a graph with 6 vertices and each vertex has degree 3, then it has

- a) 3 edges
- b) 6 edges
- c) 12 edges
- d) 9 edges

Consider the wheel graph W_4 described in the picture below. Then which of the following walk is a cycle?



- a) 0, 1, 2, 0, 3, 4, 3
- b) 0,1,2,0
- c) 0, 1, 2, 3, 4
- d) 0,1,2,1,0

Answer:b)

Consider the lattice D_{18} , the divisors of 18 ordered by divisibility. Then the least upper bound of $\{2, 3\}$ in D_{18} is.

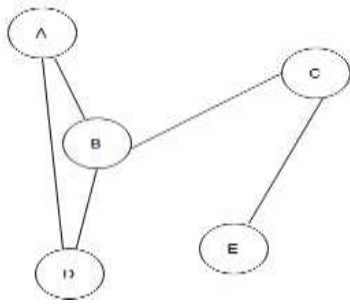
- a) 1
- b) 6
- c) 12
- d) 18

For a given graph G having v vertices and 10 edges which is connected and has no cycles, which of the following is true?

- a) $v=10$
- b) $v=11$
- c) $v=9$
- d) $v=5$

Answer: b)

. For the below graph, which of the following is/are the cut vertices?



- a) B and E b) C and D
c) A and E d) C and B

Answer: d)

A graph is said to be planar if it can be drawn in a plane so that.....

- (a) No edges cross (b) Edges cross
(c) Both (a) and (b) (d) None of these

The connected components of the given graph are

- (a) {a, b, c}, {d, e, f}, {g, h, i}, {j} (b) {a, c}, {d, e, f}, {g, h, i}, {j}
(c) {a, b, c}, {d, f}, {g, h, i}, {j} (d) {a, b, c}, {d, e, f}, {g, h, i}

A complete graph K_n is regular of degree.....

- (a) $n + 1$ (b) $2n - 1$
(c) $2n + 1$ (d) $n - 1$

What is the Boolean expression for a three input AND gate?

- (a) $A \cdot B \cdot C$ (b) $A + B + C$
(c) $A' \cdot B' \cdot C'$ (d) None of these

Ans (a) $A \cdot B \cdot C$

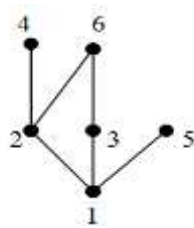
Which of the following is a sub lattice of lattice D_{36} ?

- (a) D_{12} (b) D_{16}
(c) D_{15} (d) D_{14}

The number of atoms in a Boolean algebra with n variables is:

- (a) n (b) 2^n (c) $2^n - 1$ (d) Depends on structure

Which of the following is neither maximal nor a minimal element of the given Hasse diagram?



- a) 5 b) 2
c) 1 d) 4

A graph has 25 edges. The sum of the degrees of all the vertices of this graph is:

- (A) 100 (B) 25 (C) 75 (D) 50

The maximum degree of any vertex in a simple graph with 10 vertices

- (A) 10 (B) 5 (C) 20 (D) 9

The sum of degrees of all vertices of a graph is:

- (A) a prime number (B) an even number
(C) any natural number (D) a perfect square

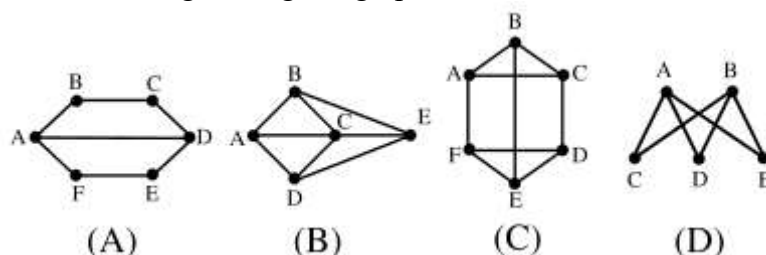
The number of odd vertices in any graph is:

- (A) odd (B) even (C) multiple of three (D) none

If a simple graph G contains exactly one edge between each pair of distinct vertices, then G is:

- (A) null graph (B) complete graph (C) regular graph (D) platonic graph

Identify which of the following is a regular graph.



The total number of edges in a complete graph with n vertices is:

- (A) n (B) $n^2 - 1$ (C) $\frac{n}{4}$ (D) $\frac{n(n-1)}{2}$

The number of edges in a complete graph K_5 is:

- (A) 10 (B) 15 (C) 25 (D) 7

A simple graph with k -components and n vertices can have at most ofedges.

- (a) $(n - k)(n - k + 1)/2$ (b) $(n + k)(n - k + 1)/2$
 (c) $(n - k)(n + k + 1)/2$ (d) $(n - k)(n - k - 1)/2$

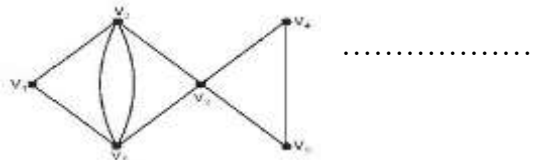
The graph given is Circuit.



- (a) Non-Euler circuit (b) Non-Hamiltonian
 (c) Both Euler and Hamiltonian (d) None of these

Ans (c) Both Euler and Hamiltonian

The given graph is



- (a) Euler circuit but not a Hamiltonian circuit (b) Hamiltonian circuit but not a Euler circuit
 (c) Both Euler and Hamiltonian (d) None of these

Ans (a) Euler circuit but not a Hamiltonian circuit

A complete graph K_n is regular of degree.....

- (a) $n + 1$ (b) $2n - 1$
 (c) $2n + 1$ (d) $n - 1$

A Boolean function is a function that:

- (a) Takes real values (b) Takes only values 0 or 1
 (c) Takes integer values (d) Takes complex values

Answer: B

If $p \rightarrow q$ is a conditional statement, then contrapositive would be

- (a) $q \rightarrow p$ (b) $p \rightarrow \sim q$
(c) $\sim q \rightarrow \sim p$ (d) $\sim p \rightarrow q$

The proposition $(p \wedge q) \rightarrow (p \vee q)$ is

- a) Tautology b) Contradiction
c) Fallacy d) None of the above
(a)

The number of edges in the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 5, 10, 15, 30\}$ is ____

- a) 5 b) 8
c) 9 d) 6

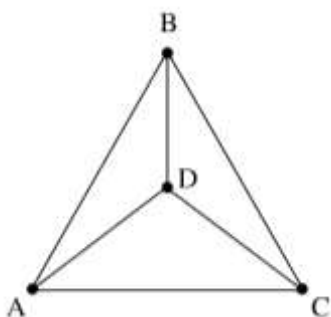
Answer: c

. Let L be a lattice. Then $a \wedge b = a$ if and only if

- a) $a \vee b = b \vee a$ b) $a \vee b = b$
c) $a \wedge b = b \wedge a$ d) None of these

Answer: b

Which of the following is the correct adjacency matrix M_A of the graph G shown below:



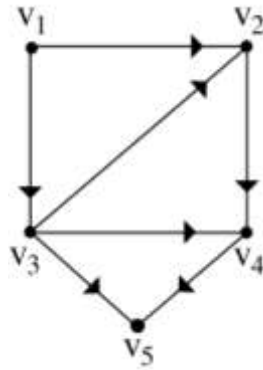
$$(A) M_A = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$(B) M_A = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$(C) M_A = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$(D) M_A = \begin{matrix} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

Consider the directed graph G shown below and determine which of the following is the appropriate adjacency matrix M_A .



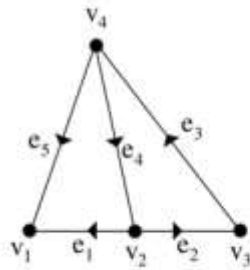
$$(A) M_A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$(B) M_A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$(C) M_A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

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Consider the directed graph G shown below and determine which of the following is the appropriate incidence matrix M_I . □



$$(A) M_I = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} -1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$(B) M_I = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} -1 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$(C) M_I = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} -1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$(D) M_I = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} -1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \end{bmatrix} \end{matrix}$$